

ME1103

Principles of Mechanics and Materials

REVIEW ON FORCE AND MOMENT

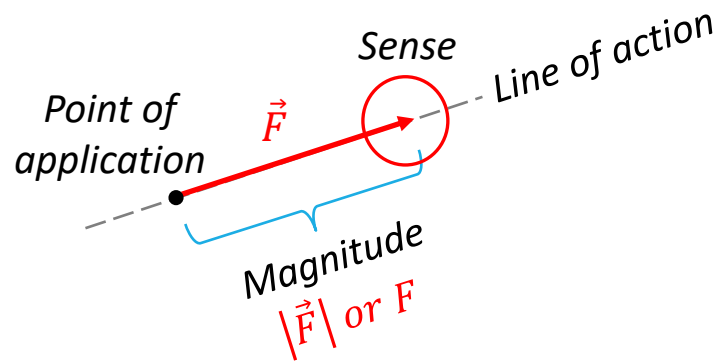
Learning Outcomes

- Understand the definition of a vector, in relation to position vector, force and moment.
- Understand the parallelogram law and triangle rule for adding vectors.
- Able to resolve a vector into its components in the rectangular coordinate system.
- Understand the moment of a force and its effect.
- Understand couple which is the moment of a pair of equal and opposite forces.
- Understand the concept of replacing a system of forces by an equivalent force-moment pair.

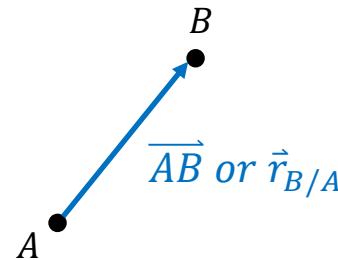
Vectors in Mechanics

- Vectors in physics are physical quantities that have both magnitude and direction.
- They are often the primary quantities to be evaluated in the study of mechanics, such as position vector, displacement, velocity, acceleration, moment, force, and etc.

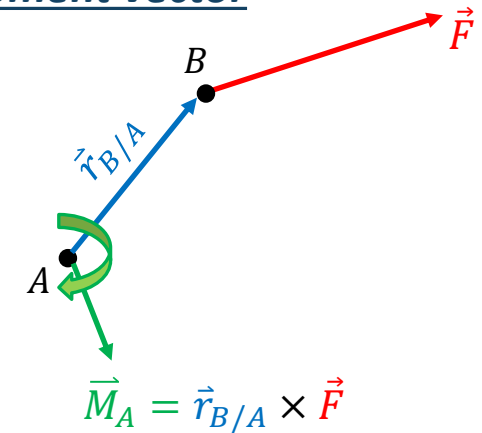
Force vector



Position vector



Moment vector

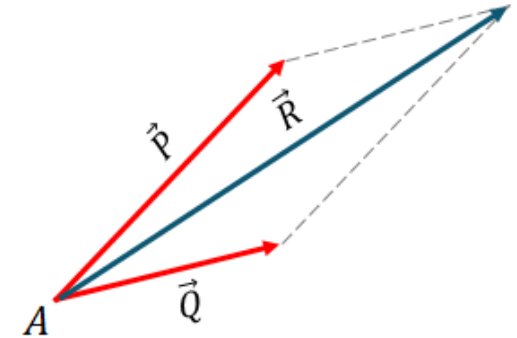


Addition of Vectors

- It is shown experimentally that the addition of two forces can be represented by a single resultant force, which is equivalent to the diagonal of a parallelogram formed with the two vectors as the adjacent sides.

$$\vec{R} = \vec{P} + \vec{Q}$$

This is called the parallelogram law of vector addition.



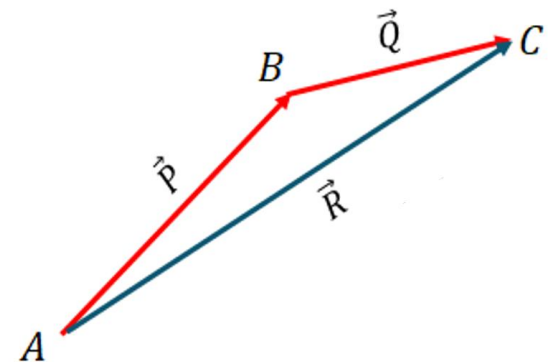
- By shifting $\vec{Q}$ to the end of $\vec{P}$, vector addition can also be expressed as a force triangle, which is known as the triangle rule for vector addition.

Cosine rule:

$$R^2 = P^2 + Q^2 - 2PQ\cos B$$

Sine rule:

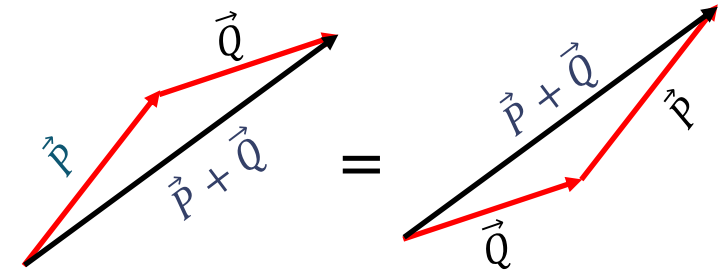
$$\frac{\sin A}{Q} = \frac{\sin B}{R} = \frac{\sin C}{P}$$



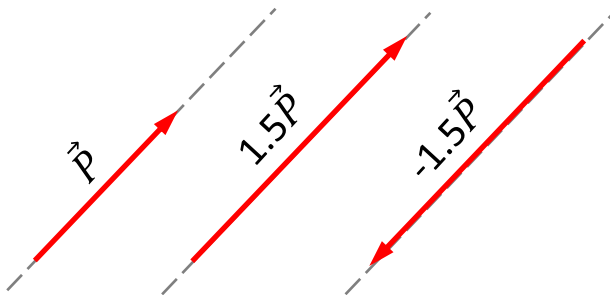
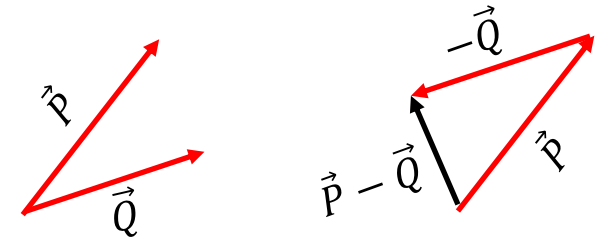
Addition of Vectors...

- Vector addition is a commutative operation, that is, the order of addition is not important.

$$\vec{P} + \vec{Q} = \vec{Q} + \vec{P}$$



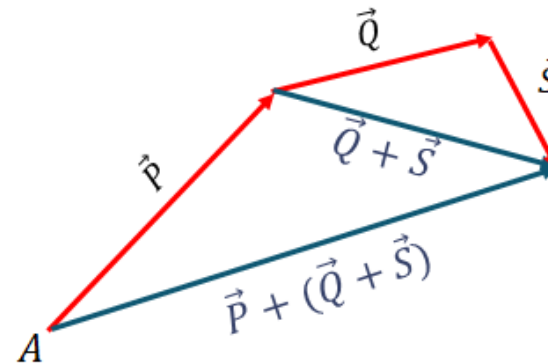
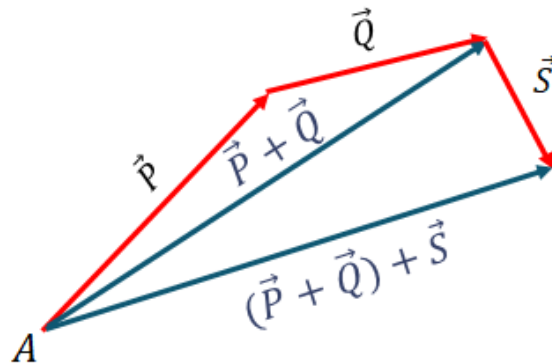
- Vector subtraction of two vectors ($\vec{P} - \vec{Q}$) is the addition of the second vector with opposite sense.



- Multiplication of a vector by a scalar changes its magnitude and maybe its sense if the scalar value is a negative, but not the line of action.

Addition of Vectors...

- To add three or more vectors, we can use the triangle rule repeatedly as shown



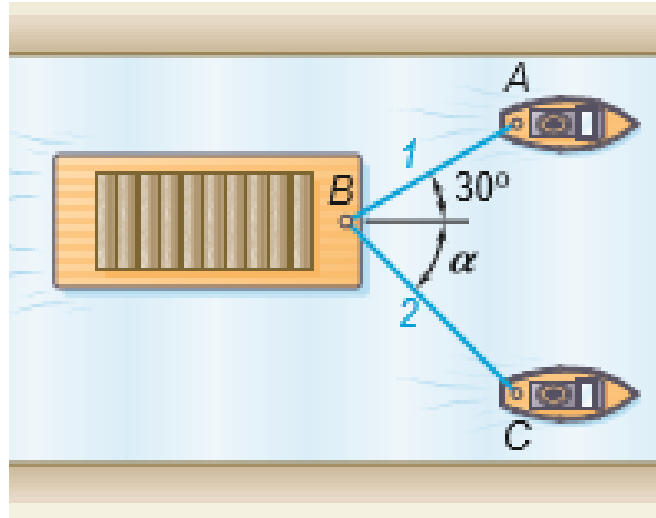
- Vector addition is also an associative operation, which again states that the order of addition is not important.

$$\vec{P} + \vec{Q} + \vec{S} = (\vec{P} + \vec{Q}) + \vec{S} = \vec{P} + (\vec{Q} + \vec{S})$$

- Being commutative and associative, means that the order of addition is immaterial.

$$\vec{P} + \vec{Q} + \vec{S} = \vec{S} + \vec{Q} + \vec{P} = \vec{P} + \vec{S} + \vec{Q} = \dots$$

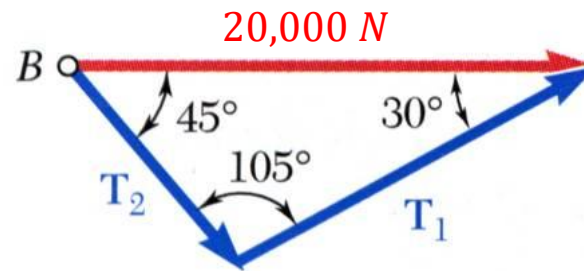
Example 1.1



A barge is pulled by two tugboats. If the resultant of the forces exerted by the tugboats is a 20 kN directed along the axis of the barge, determine

- (a) the tension in each of the ropes knowing that $\alpha = 45^\circ$, and
- (b) the value of α for which the tension in rope 2 is minimum, and this corresponding value.

Example 1.1...



(a) Based on sine law,

$$\frac{T_1}{\sin 45^\circ} = \frac{T_2}{\sin 30^\circ} = \frac{20 \text{ kN}}{\sin 105^\circ}$$

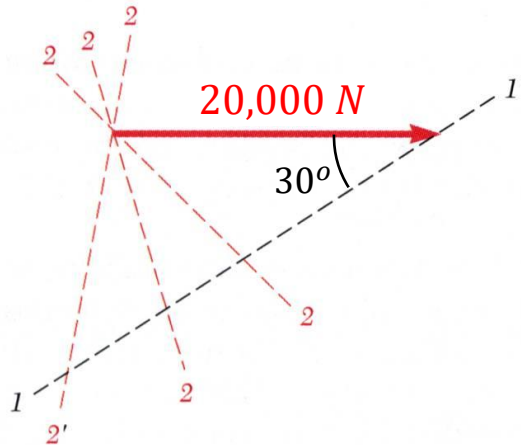
Hence

$$\Rightarrow T_1 = \frac{20 \text{ kN}}{\sin 105^\circ} (\sin 45^\circ) = \mathbf{14.64 \text{ kN}}, \quad \text{and}$$

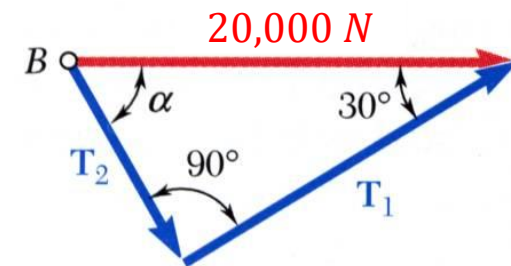
$$T_2 = \frac{20 \text{ kN}}{\sin 105^\circ} (\sin 30^\circ) = \mathbf{10.35 \text{ kN}}$$

Example 1.1...

(b) The tension in rope 2 varies with the angle α .



And based on observation, the minimum tension in rope 2 occurs when the line of actions of the two forces is perpendicular.



That is,

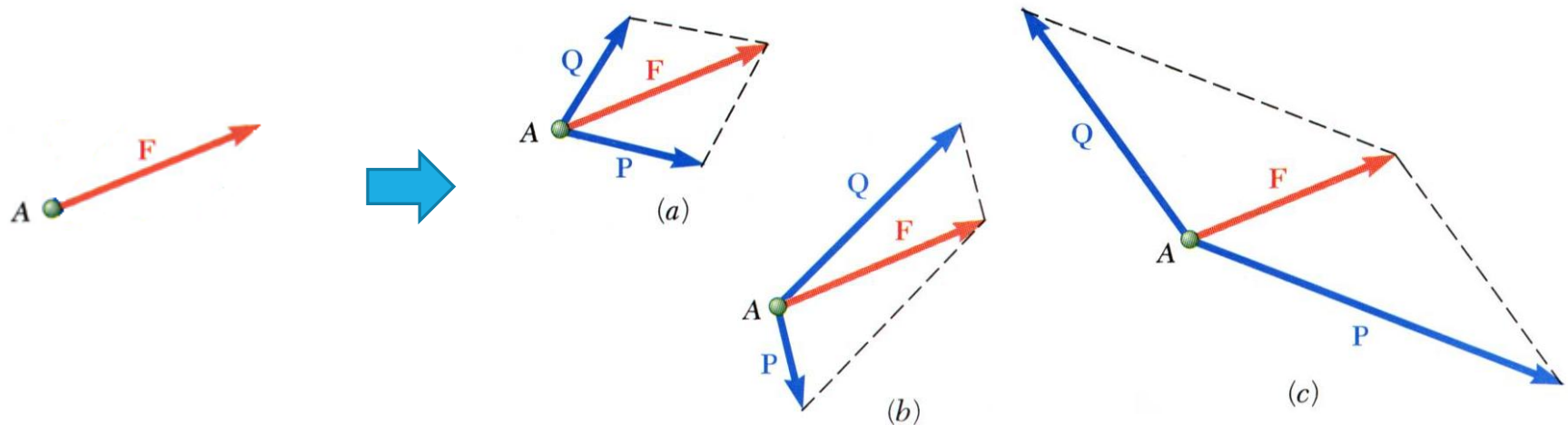
$$\alpha = 60^\circ$$

Based on this configuration,

$$\frac{T_2}{\sin 30^\circ} = \frac{20 \text{ kN}}{\sin 90^\circ} \Rightarrow T_2 = 10 \text{ kN}$$

Resolving Vector into Components

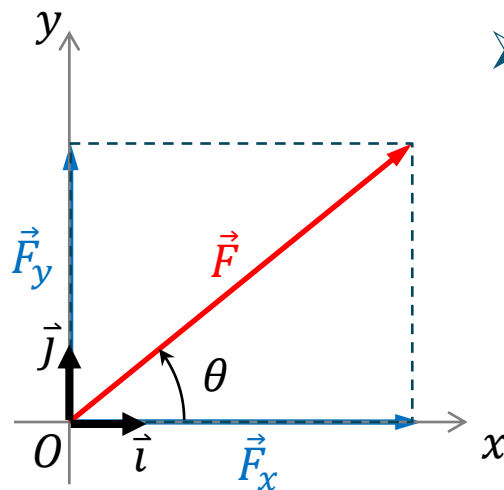
- Given that two vectors can be added to give a resultant vector, using either the parallelogram law or triangle rule.
- The reverse can be said about a vector, which can be decomposed into two or more components, such as...



- There are infinite possible ways to decompose a vector.
- And getting the magnitudes of the components are not straightforward...unless you associated the vector with an orthogonal coordinate system.

Resolving Vector into Components...

- One such coordinate system is the rectangular (cartesian) coordinate system, which is defined by lines of action that are perpendicular to each other, such as in the horizontal and vertical directions for 2D space:



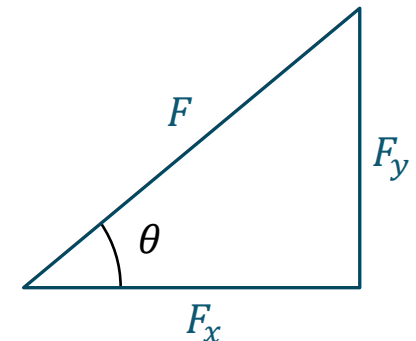
- With the coordinate system, vector $\vec{F}$ can be uniquely decomposed into its rectangular vector components, as

$$\vec{F} = \vec{F}_x + \vec{F}_y = F_x \vec{i} + F_y \vec{j} = \begin{Bmatrix} F_x \\ F_y \end{Bmatrix}$$

where $\vec{F}_x$ and $\vec{F}_y$ are along the x - and y - axis, respectively, and $\vec{i}$ and $\vec{j}$ are the unit vectors (with magnitude = 1) along the two axes.

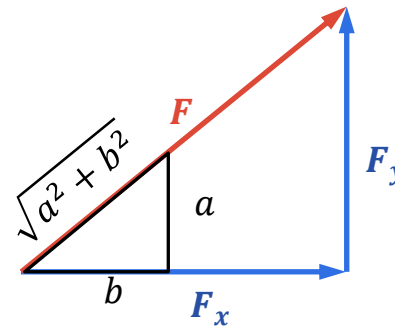
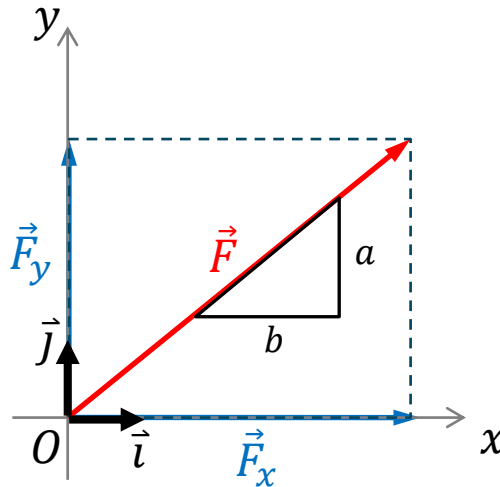
- Defining the angle θ of the force, which is measured from the x-axis in the counter-clockwise direction, the magnitude and direction of the components are given by

$$F_x = F \cos \theta \quad \text{and} \quad F_y = F \sin \theta$$



Resolving Vector into Components...

- Suppose you are given a right-angled triangle as shown.



The right-angled triangle and the force triangle are similar triangles.

- Hence, the force can also be decomposed into its component as follows.

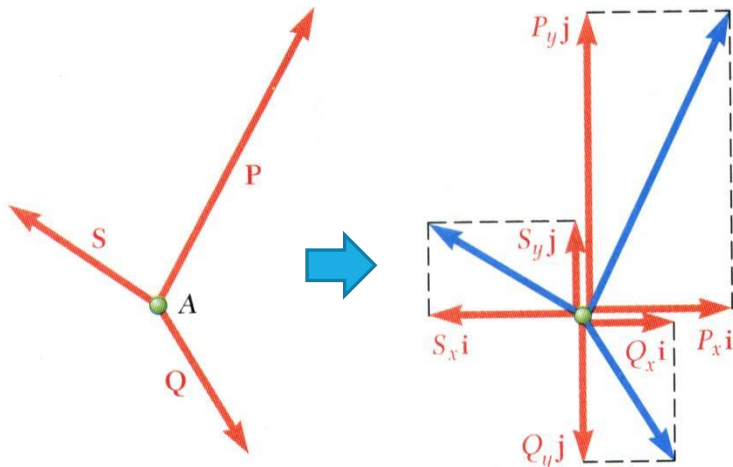
$$\frac{F_x}{F} = \frac{b}{\sqrt{a^2 + b^2}} \Rightarrow F_x = \left(\frac{b}{\sqrt{a^2 + b^2}} \right) F$$

And similarly for the y-component

$$F_y = \left(\frac{a}{\sqrt{a^2 + b^2}} \right) F$$

Addition Vectors by Summing Components

- With the understanding of vector decomposition in orthogonal system, we can perform vector addition of many vectors simply by summing their components.



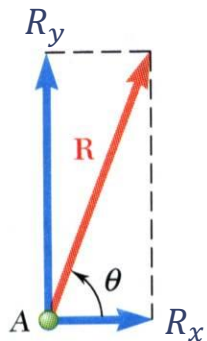
- The vector sum of the three vectors is

$$\vec{R} = \vec{P} + \vec{Q} + \vec{S}$$

$\swarrow \quad \quad \quad \searrow \quad \quad \quad \swarrow$
 $P_x \vec{i} + P_y \vec{j} \quad \quad Q_x \vec{i} + Q_y \vec{j} \quad \quad S_x \vec{i} + S_y \vec{j}$

- Adding the components separately, we have

$$R_x = P_x + Q_x + S_x; \text{ and } R_y = P_y + Q_y + S_y$$



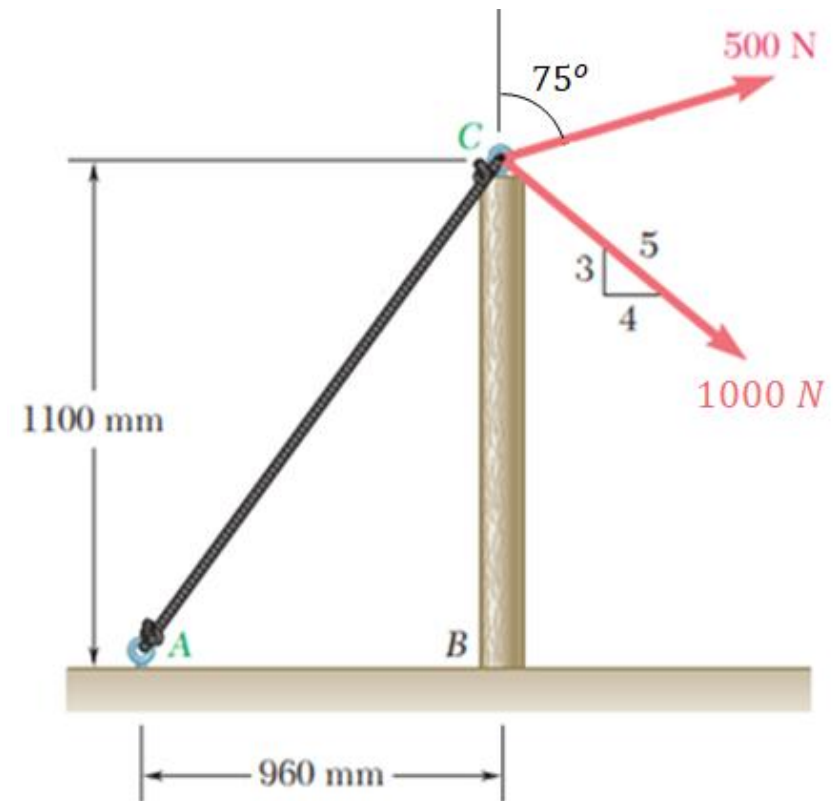
- Finally, to express the resultant vector in magnitude and direction form,

$$R = \sqrt{R_x^2 + R_y^2}, \quad \text{and} \quad \theta = \tan^{-1} \left(\frac{R_y}{R_x} \right)$$

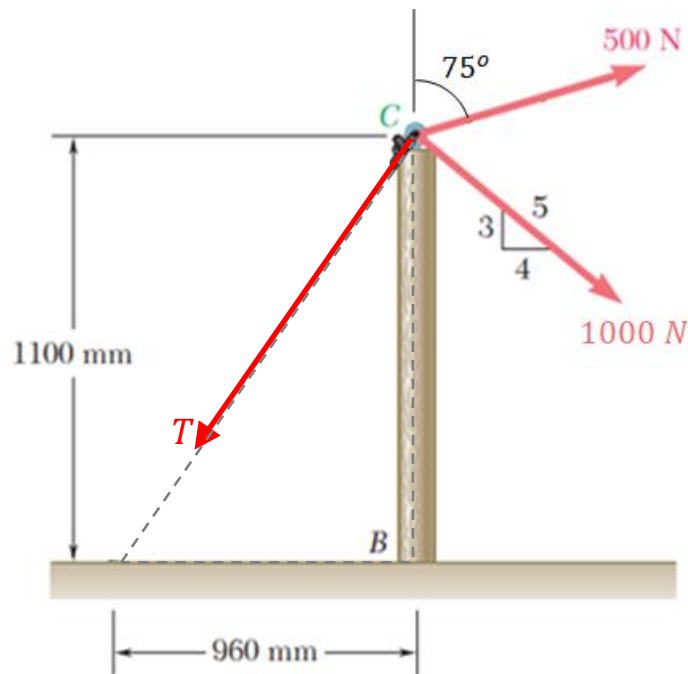
Example 1.2

A vertical pole is supported by a rope and loaded at point *C* as shown.

- (a) Determine the tension in rope *AC* if the resultant of the three forces exerted at point *C* is to be vertical.
- (b) Suppose if the pole can only withstand a maximum compressive load of 4000 *N*, can it survive the loading condition in (a) with a safety factor of 2?



Example 1.2...



For the 500 N force,

$$F_{500,x} = 500 \sin 75 = 483 \text{ N}$$

$$F_{500,y} = 500 \cos 75 = 129.4 \text{ N}$$

For the 1000 N and based on similar triangles,

$$F_{1000,x} = 1000 \left(\frac{4}{5} \right) = 800 \text{ N};$$

$$F_{1000,y} = 1000 \left(\frac{-3}{5} \right) = -600 \text{ N}$$

The length of the cable is $\sqrt{0.96^2 + 1.1^2} = 1.46 \text{ m}$

With this, the tension in the cable can be decomposed as

$$T_x = \frac{-0.96}{1.46} T = -0.6575T; \text{ and } T_y = \frac{-1.1}{1.46} T = -0.7534T$$

Example 1.2...

(a) The resultant of the three forces at C is

$$R_x = -0.6575T + 483 + 800 = -0.6575T + 1283$$

$$R_y = -0.7534T + 129.4 - 600 = -0.7534T - 470.6$$

For the resultant to be vertical, this implies that $R_x = 0$

$$-0.6575T + 1283 = 0 \Rightarrow T = \mathbf{1951.3\text{ N}}$$

(b) Substituting T into the force equation in y - direction, gives

$$R_y = -0.7534(1951.3) - 470.6 = \mathbf{-1940.7\text{ N}}$$

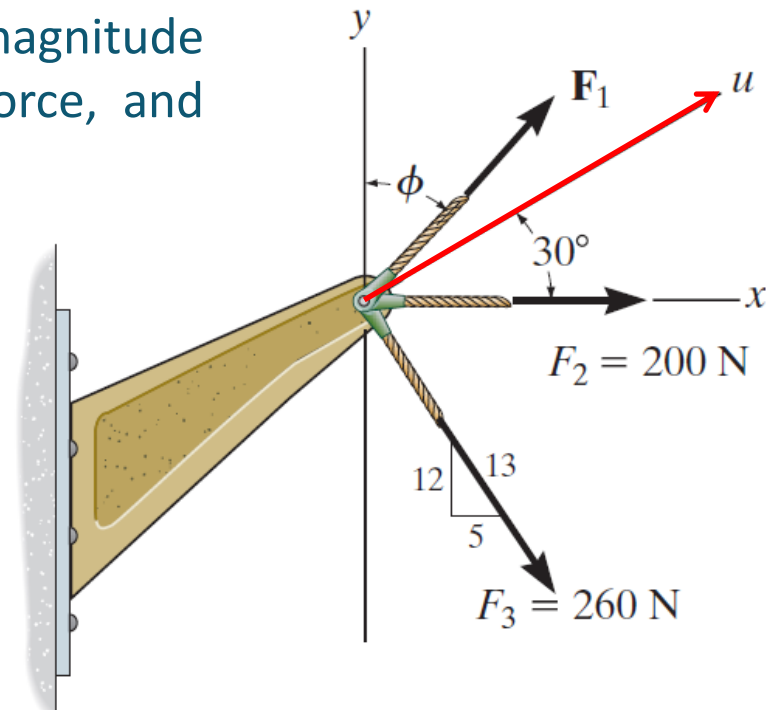
To have a safety factor of 2, the limiting load on the pole is only $4000/2 = 2000\text{ N}$.

Since the compressive load is less than the load limit of 2000 N , the pole is safe to be used under the given loading conditions.

Example 1.3

A bracket is subjected to the forces as shown in the figure.

- Given that the magnitude of $\vec{F}_1$ is 500 N. What should ϕ be so that the resultant force is in the direction $\vec{u}$.
- Suppose ϕ is fixed at 30° . Determine the magnitude of F_1 to give rise to the least resultant force, and hence this corresponding resultant force.



Example 1.3...

(a) Resolving the forces into the x- and y- components,

$$(F_1)_x = 500\sin\phi; (F_1)_y = 500\cos\phi$$

$$(F_2)_x = 200 \text{ N}; (F_2)_y = 0$$

$$(F_3)_x = 260\left(\frac{5}{13}\right) = 100 \text{ N}; (F_3)_y = 260\left(\frac{-12}{13}\right) = -240 \text{ N}$$

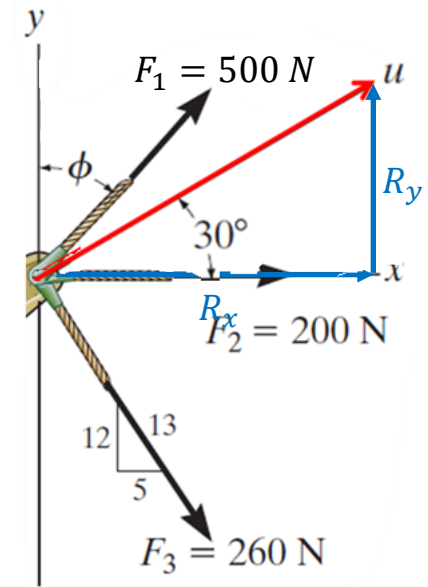
With these, the resultant force is

$$R_x = 500\sin\phi + 300; \quad R_y = 500\cos\phi - 240$$

For the resultant force to be in the $\vec{u}$ direction, the force ratio is $\frac{R_y}{R_x} = \tan 30^\circ$, i.e.

$$\frac{500\cos\phi - 240}{500\sin\phi + 300} = \tan 30^\circ \Rightarrow {}^\# 500\sin\phi - 866.025\cos\phi = -715.692$$

$$\Rightarrow 1000 \sin(\phi - 60^\circ) = -715.692 \Rightarrow \phi = 14.3^\circ$$

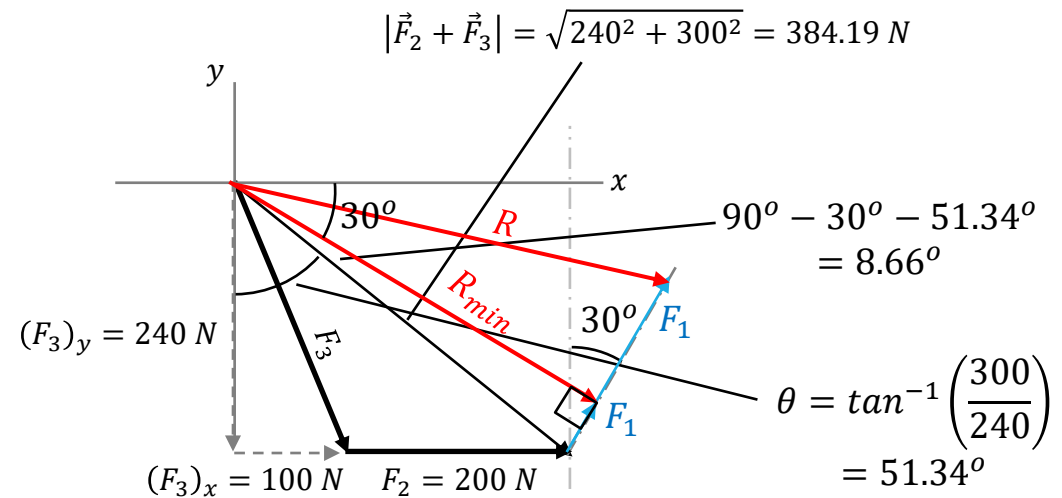
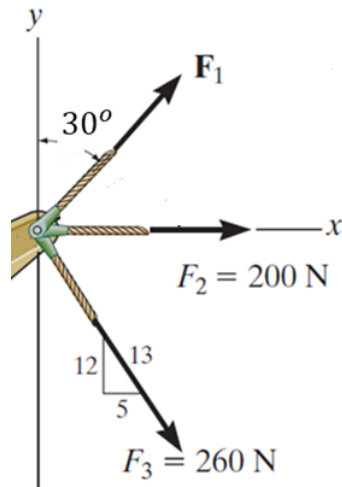


[#]R-formula: $A\sin x + B\cos x = R\sin(x + \alpha)$, where $R = \sqrt{A^2 + B^2}$ and $\alpha = \tan^{-1}(B/A)$.

Example 1.3...

(b) Using geometrical approach:

For the least resultant force $\vec{R}$ to occur, it must be perpendicular to $\vec{F}_1$. With this, the force diagram is



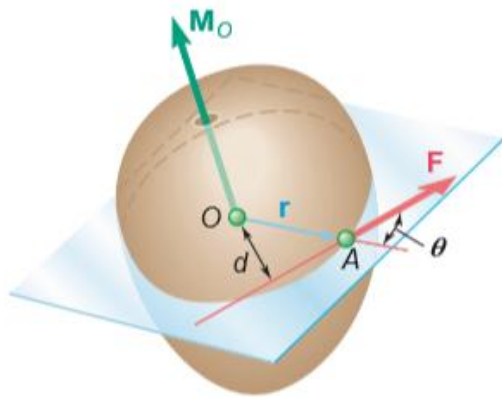
Finally, based on the right-angle triangle, we have the magnitudes

$$F_1 = 384.19 \sin(8.66) = \mathbf{57.85 \text{ N}}$$

$$R_{min} = 384.19 \cos(8.66) = \mathbf{379.8 \text{ N}}$$

Moment of a Force about a Point

- Moment of a force is a measure of its tendency to cause a body to rotate about a specific point or axis.

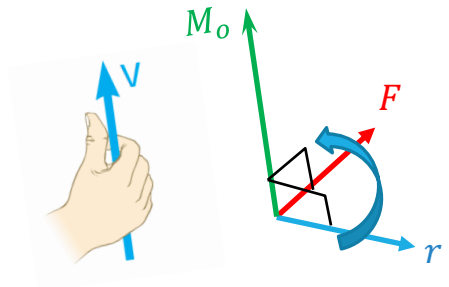


- By definition, the moment of a force F about O is defined by

$$\vec{M}_O = \vec{r} \times \vec{F}$$

where $\vec{r}$ is the relative position vector that joins point O to the force application point A. It is also referred to as the moment arm vector.

- The moment of a force is the vector product (also called cross-product) of $\vec{r}$ and $\vec{F}$. The direction of $\vec{M}_O$ must be perpendicular to the plane containing $\vec{r}$ and $\vec{F}$, which follows the right-hand screw rule.



Vector Product

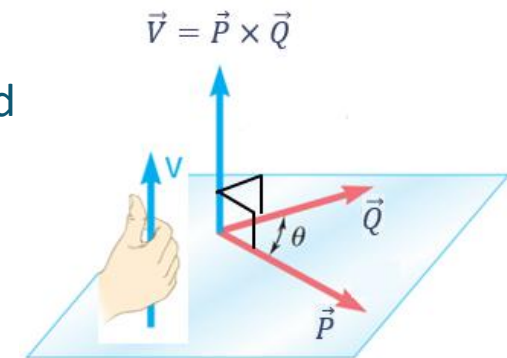
- Vector product of two vectors $\vec{P}$ and $\vec{Q}$ is defined as the vector $\vec{V}$ as follows

$$\vec{V} = \vec{P} \times \vec{Q} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ P_x & P_y & P_z \\ Q_x & Q_y & Q_z \end{vmatrix} = \begin{vmatrix} P_y & P_z \\ Q_y & Q_z \end{vmatrix} \vec{i} - \begin{vmatrix} P_x & P_z \\ Q_x & Q_z \end{vmatrix} \vec{j} + \begin{vmatrix} P_x & P_y \\ Q_x & Q_y \end{vmatrix} \vec{k}$$

$$= (P_y Q_z - P_z Q_y) \vec{i} + (P_z Q_x - P_x Q_z) \vec{j} + (P_x Q_y - P_y Q_x) \vec{k}$$

which satisfies the following conditions:

- Line of action of $\vec{V}$ is perpendicular to the plane containing $\vec{P}$ and $\vec{Q}$
- The magnitude of $\vec{V}$ is given by $|\vec{V}| = |\vec{P}||\vec{Q}|\sin\theta$
- Direction of $\vec{V}$ follows the right-hand screw rule.

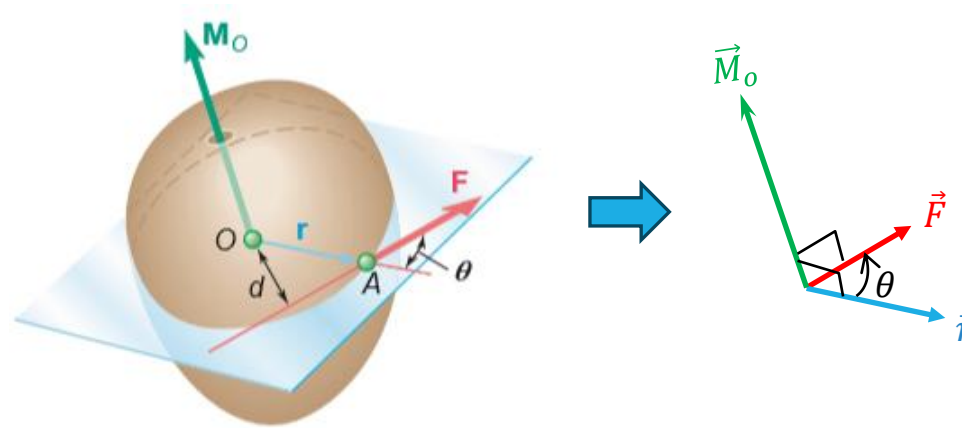


- Vector product operation is:

- **NOT** Commutative, $\vec{P} \times \vec{Q} \neq \vec{Q} \times \vec{P}$
- Distributive, $\vec{P} \times (\vec{Q} + \vec{R}) = \vec{P} \times \vec{Q} + \vec{P} \times \vec{R}$
- **NOT** Associative, $\vec{P} \times (\vec{Q} \times \vec{R}) \neq (\vec{P} \times \vec{Q}) \times \vec{R}$

Order of operations in the vector product matters.

Moment of a Force in 3D



In general, the moment of the force is given by

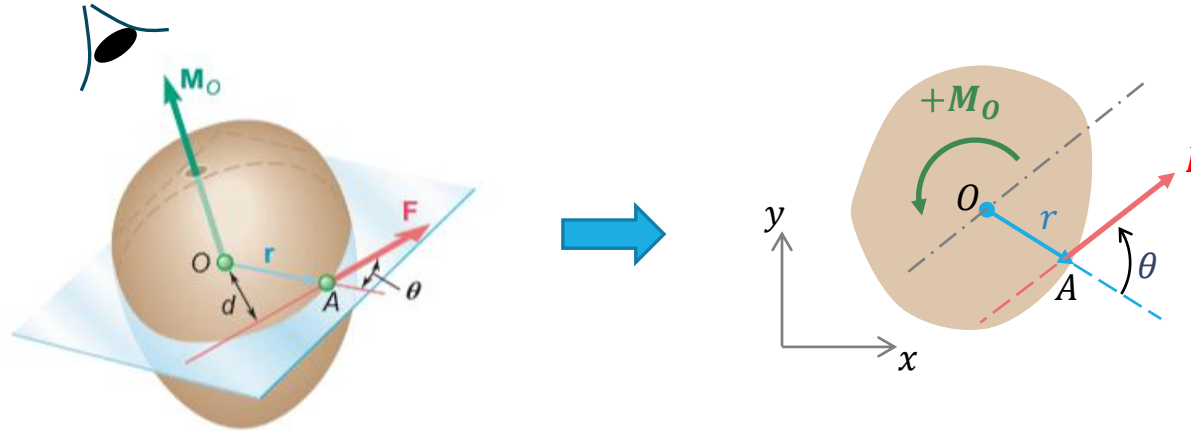
$$\vec{M} = (r_y F_z - r_z F_y)\vec{i} - (r_x F_z - r_z F_x)\vec{j} + (r_x F_y - r_y F_x)\vec{k}$$

Its magnitude can be also be evaluated as

$$|\vec{M}| = |\vec{r}||\vec{F}|\sin\theta$$

where the magnitude of a vector is $|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$.

Moment of a Force in 2D



- In 2D planar, where $\vec{r} = r_x\vec{i} + r_y\vec{j}$, and $\vec{F} = F_x\vec{i} + F_y\vec{j}$ and $r_z = F_z = 0$, the moment is reduced to

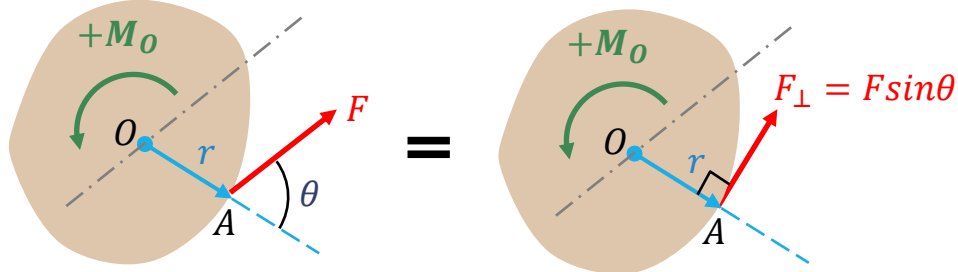
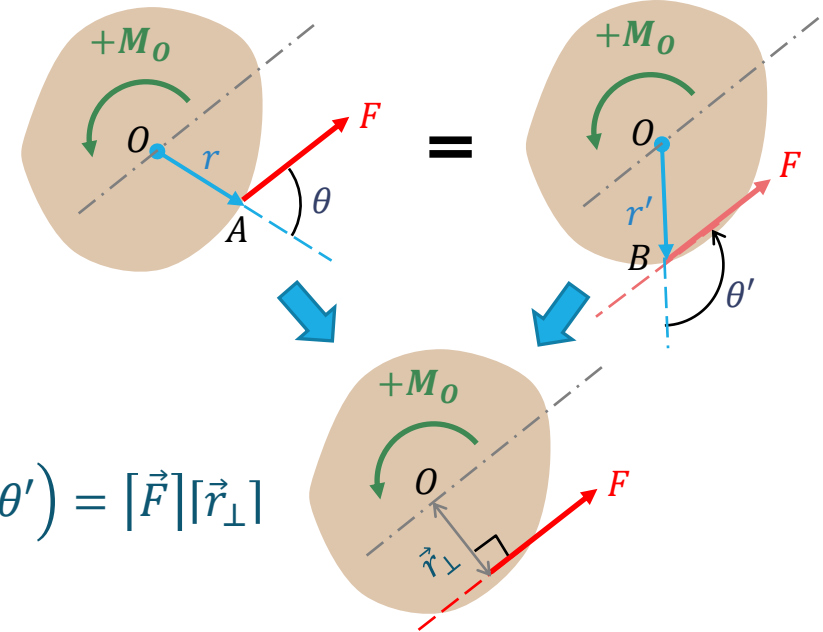
$$\vec{M}_O = (r_x F_y - r_y F_x) \vec{k}$$

- Note that the direction of the moment is always along the z-direction. Hence, moment in 2D can be regarded as a scalar quantity, where the sign of the product determines the direction of the rotational effect. Positive value implies rotational effect in the counter-clockwise direction, and vice versa.

Moment of a Force in 2D...

- Suppose the force is shifted along its line of action such that its point of action is at B . Is the moment due to this force about O changed or remain the same?
- Based on the magnitude formula,

$$M_o = |\vec{F}| |\vec{r}| \sin\theta = |\vec{F}| (|\vec{r}| \sin\theta) = |\vec{F}| (|\vec{r}'| \sin\theta') = |\vec{F}| |\vec{r}_\perp|$$



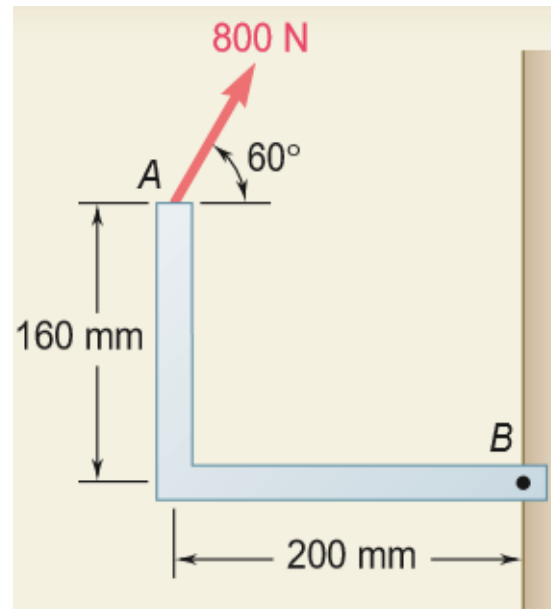
- Alternatively, we can also resolve $\vec{F}$ such that it is perpendicular to $\vec{r}$, i.e.

$$M_o = |\vec{r}| |\vec{F}| (\sin\theta) = |\vec{r}| F_\perp$$

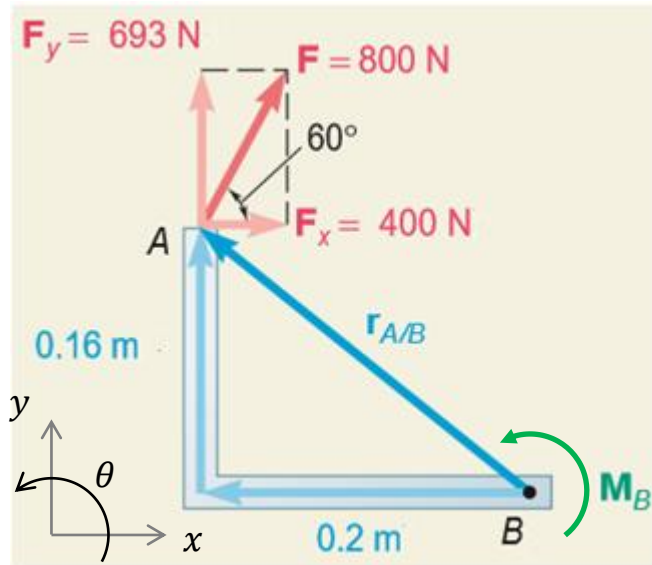
where the sense (direction) of rotation is determined by the right-hand screw rule.

Example 1.4

A force of 800 N acts on a bracket as shown. Determine the moment of the force about B .



Example 1.4...



Using the rectangular coordinate system, the force $\vec{F}$ and relative position vector $\vec{r}_{A/B}$ are:

$$\vec{r}_{A/B} = -0.2\vec{i} + 0.16\vec{j}$$

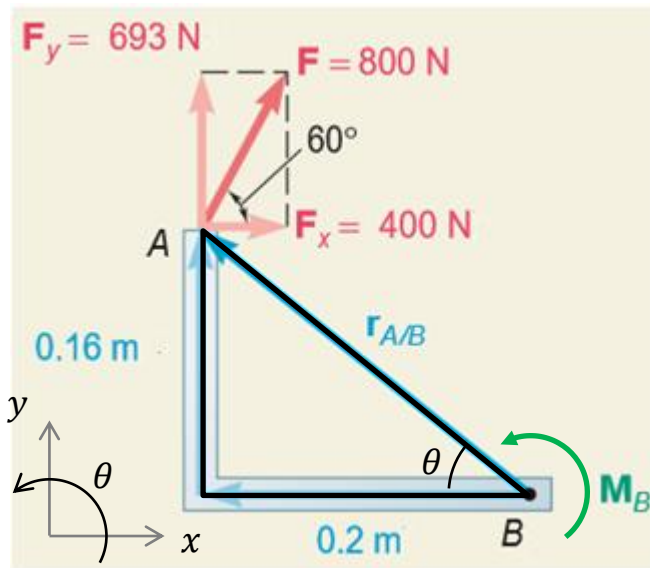
$$\vec{F} = 800\cos 60^\circ\vec{i} + 800\sin 60^\circ\vec{j} = 400\vec{i} + 693\vec{j}$$

Using the vector product formula, we get

$$M_B = x_{A/B}F_y - y_{A/B}F_x = (-0.2)(693) - (0.16)(400)$$

$$\Rightarrow M_B = -202.5 \text{ N.m}$$

Example 1.4...



Based on the right-angled triangle, we have

$$r_{A/B} = \sqrt{0.16^2 + 0.2^2} = 0.2561$$

And angle θ can be calculated as

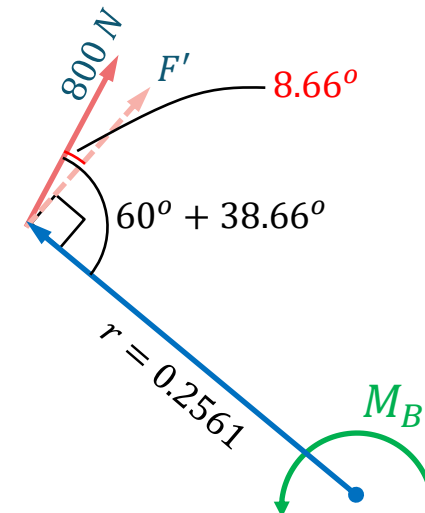
$$\tan\theta = \frac{0.16}{0.2} \Rightarrow \theta = 38.66^\circ$$

Resolving $\vec{F}$ to be perpendicular to $\vec{r}_{A/B}$, i.e.

$$F' = 800\cos(8.66) = 790.9\text{ N}$$

And the moment M_B is

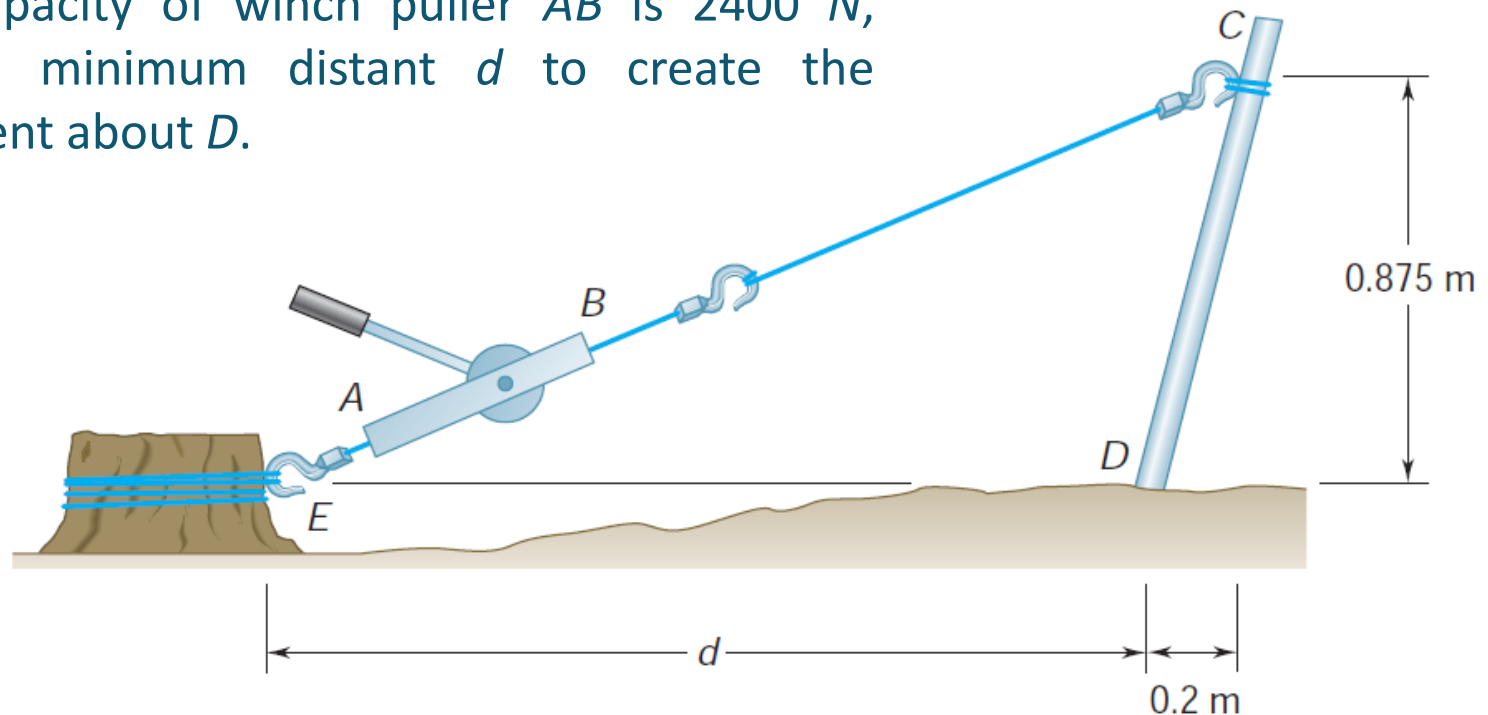
$$M_B = F'r = -790.9(0.2561) = -202.5\text{ N}$$



Example 1.5

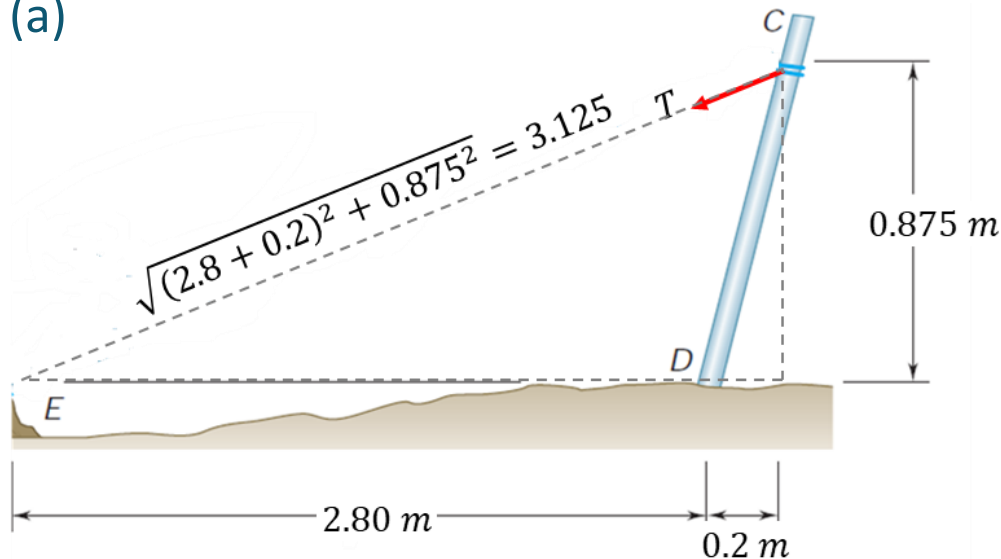
It is known that a moment of $960 \text{ N}\cdot\text{m}$ about D is required to straighten the fence post CD .

- If $d = 2.80 \text{ m}$, determine the tension that must be developed in the cable of winch puller AB to create the required moment.
- If the load capacity of winch puller AB is 2400 N , determine the minimum distant d to create the specified moment about D .



Example 1.5...

(a)



Based on geometry, the tension in the cable is

$$\begin{aligned}\vec{T} &= -\left(\frac{3.0}{3.125}T\right)\vec{i} - \left(\frac{0.875}{3.125}T\right)\vec{j} \\ &= -0.96T\vec{i} - 0.28T\vec{j}\end{aligned}$$

And the moment arm $\vec{CD}$ is

$$\vec{CD} = 0.2\vec{i} + 0.875\vec{j}$$

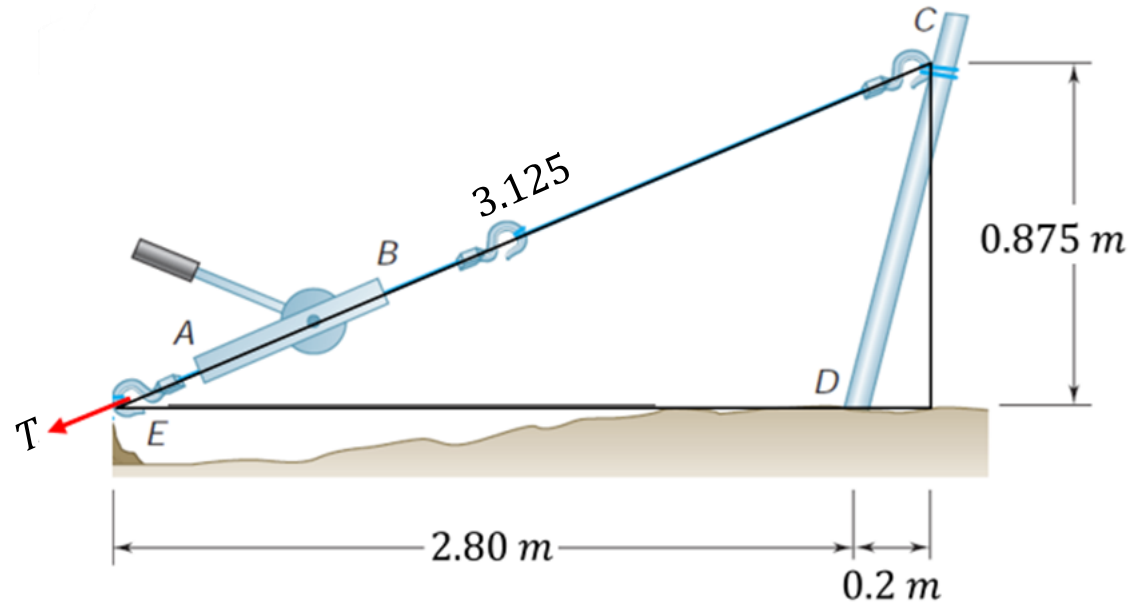
With this, the moment of the tension $\vec{T}$ induces about D is

$$M_D = CD_x T_y - CD_y T_x = (0.2)(-0.28T) - (0.875)(-0.96T) = 0.784T$$

Then matching this to the required moment of 960 *N.m*, gives

$$0.784T = 960 \Rightarrow T = \mathbf{1224.5\text{ N}}$$

Example 1.5...



Alternatively, we can move the cable tension to E . In this case, the moment of the tension about D becomes,

$$M_D = T_x(0) + T_y(2.8) \Rightarrow \left(\frac{0.875}{3.125} T \right) (2.8) = 960$$

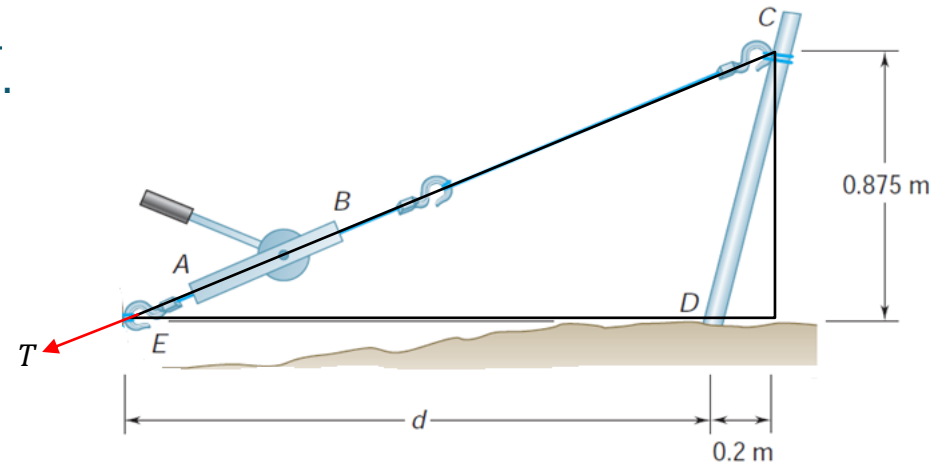
$$\Rightarrow T = 1224.5 \text{ N}$$

Example 1.5...

(b) With varying d , $CE = \sqrt{(d + 0.2)^2 + (0.875)^2}$.

And the moment of $\vec{T}$ about D becomes

$$M_D = \frac{0.875T}{\sqrt{(d + 0.2)^2 + (0.875)^2}} d$$



When the tension is at its maximum of 2400 N, this moment becomes

$$M_D = \frac{0.875(2400)}{\sqrt{(d + 0.2)^2 + (0.875)^2}} d = \frac{2100d}{\sqrt{(d + 0.2)^2 + (0.875)^2}}$$

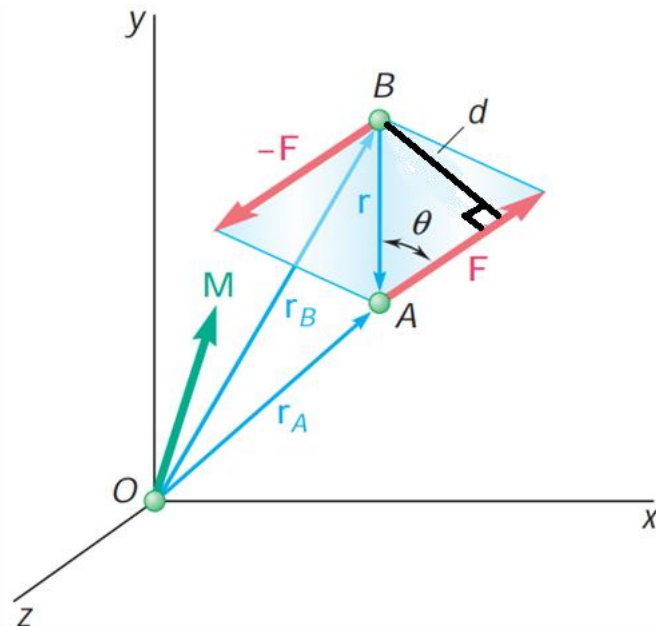
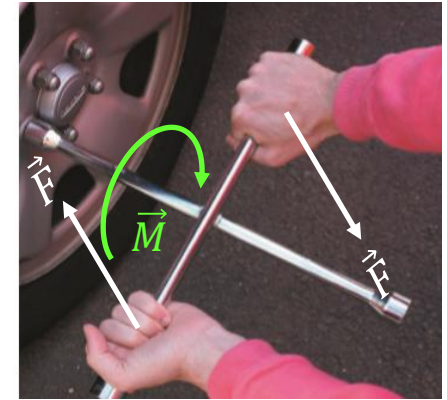
Again, matching this to the required moment, we get

$$\frac{2100d}{\sqrt{(d + 0.2)^2 + (0.875)^2}} = 960 \Rightarrow 3.7852d^2 - 0.40d - 0.8056 = 0$$

Upon solving the quadratic equation, we get $d = 0.517 \text{ m}$.

Moment of a Couple

- Two forces $\vec{F}$ and $-\vec{F}$ having the same magnitude, with parallel lines of action, but opposite sense are said to form a couple.
- Couple is a special type of moment that produces pure rotational effect on a body.



The moment due to a couple about O is

$$\vec{M} = \vec{r}_A \times \vec{F} + \vec{r}_B \times (-\vec{F}) = (\vec{r}_A - \vec{r}_B) \times \vec{F} = \vec{r}_{A/B} \times \vec{F}$$

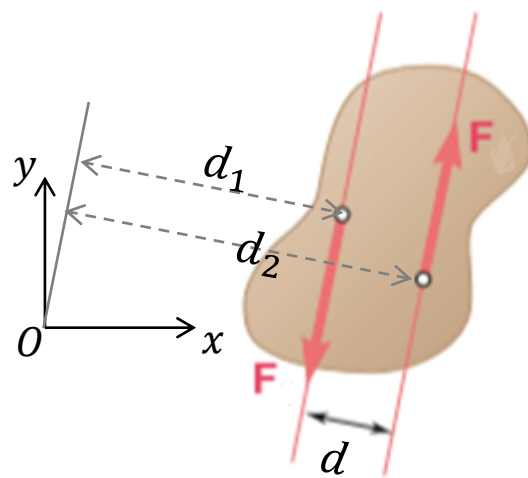
where its magnitude is

$$M = rF \sin \theta = Fd$$

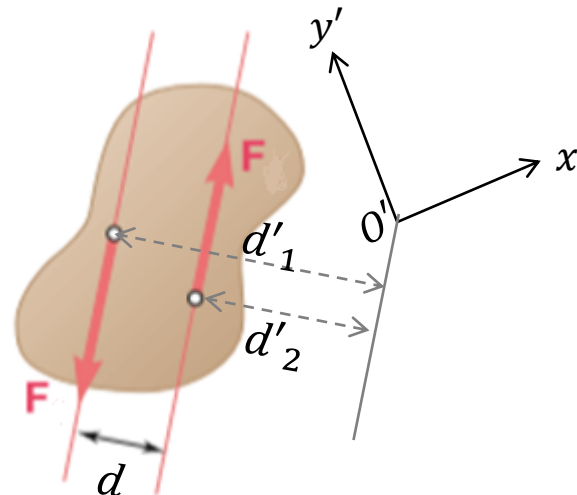
and d is the shortest (perpendicular) distance between the pair of forces, and its sense is defined by the right-hand rule.

Moment of a Couple...

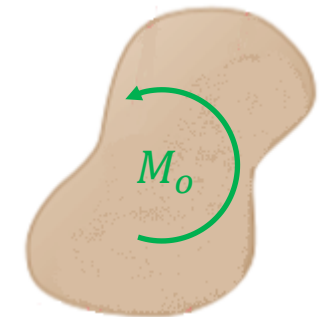
- The moment of the couple is independent of the choices of the coordinate systems. This means that it is a free vector that produces the same rotational effect w.r.t. any point of reference.



$$\begin{aligned} M_O &= -Fd_1 + Fd_2 \\ &= F(d_2 - d_1) \\ &\Rightarrow M_O = Fd \end{aligned}$$



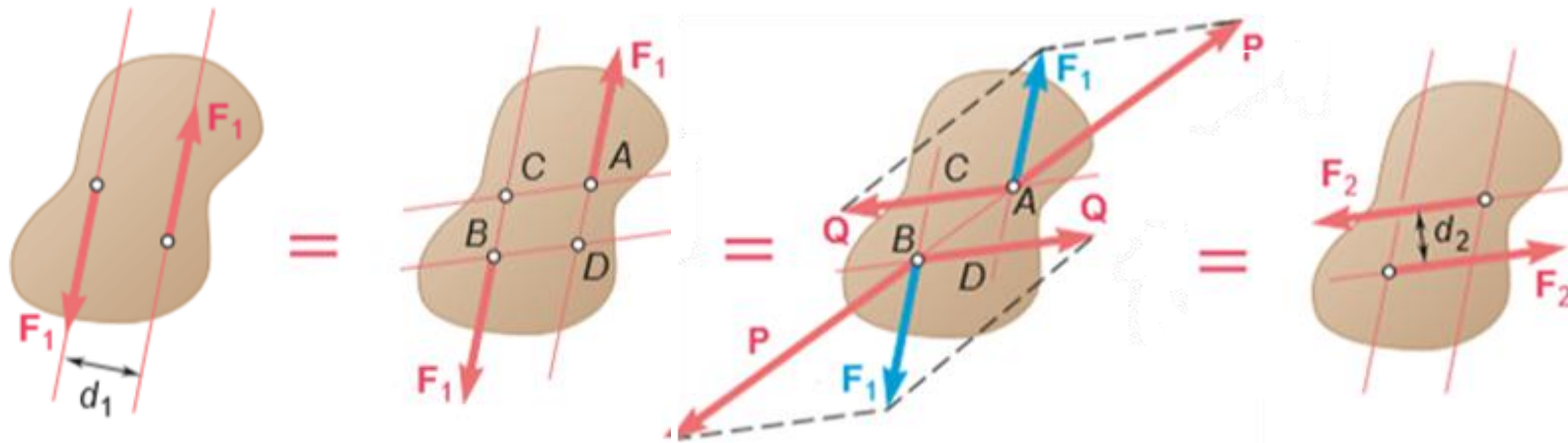
$$\begin{aligned} M_{O'} &= Fd'_1 - Fd'_2 \\ &= F(d'_1 - d'_2) \\ &\Rightarrow M_{O'} = Fd \end{aligned}$$



Placement of a couple M_O is not important because it produces the same rotational effect about any moment point.

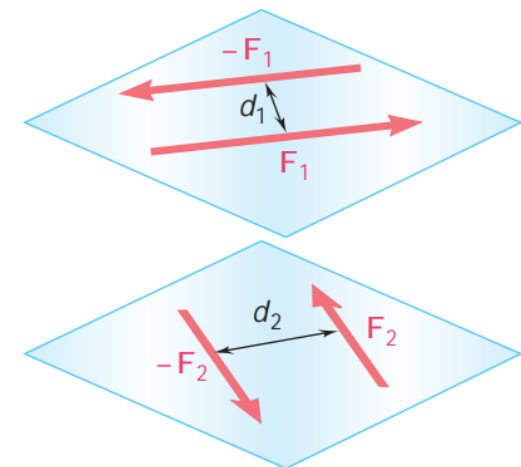
Moment of a Couple...

- A couple can also be represented by infinite force pairs.



- Two couples are identical if:

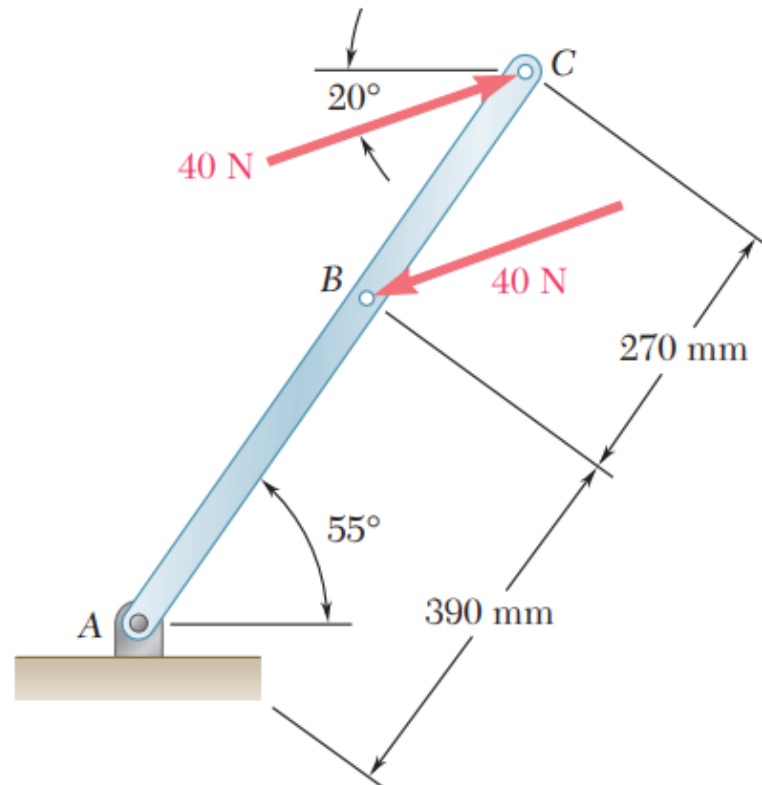
- they have same magnitudes, $F_1 d_1 = F_2 d_2$
- the two couples lie in parallel planes, and
- the two couples have the same sense or the tendency to cause rotation in the same direction.



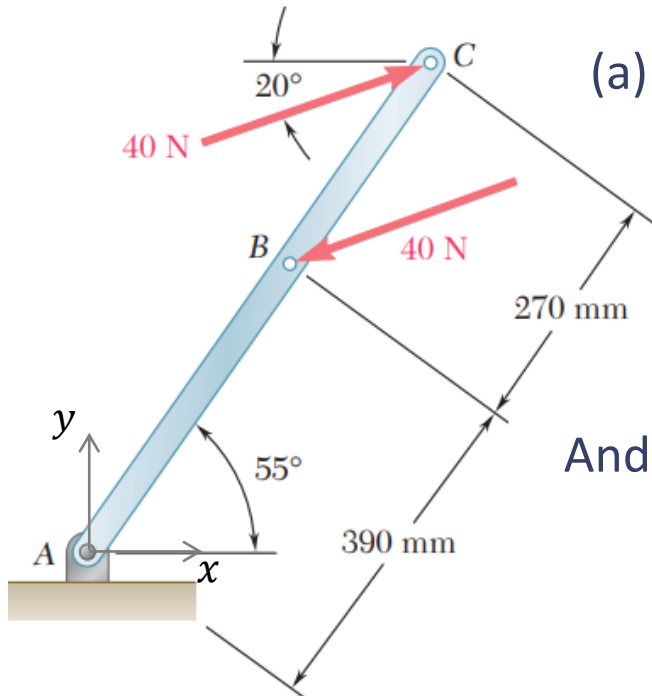
Example 1.6

Two parallel 40-N forces are applied to a lever as shown. Determine:

- The sum of the moments of the two forces about point A .
- The moment of the couple induced about B and C .



Example 1.6...



(a) The position vectors $\vec{r}_{B/A}$ and $\vec{r}_{C/A}$ are given by:

$$\vec{r}_{B/A} = 0.39\cos 55^\circ \vec{i} + 0.39\sin 55^\circ \vec{j} = 0.2237\vec{i} + 0.3195\vec{j}$$

$$\vec{r}_{C/A} = 0.66\cos 55^\circ \vec{i} + 0.66\sin 55^\circ \vec{j} = 0.3786\vec{i} + 0.5406\vec{j}$$

And the forces at B and C are:

$$\vec{F}_B = -40\cos 20^\circ \vec{i} - 40\sin 20^\circ \vec{j} = -37.59\vec{i} - 13.68\vec{j}$$

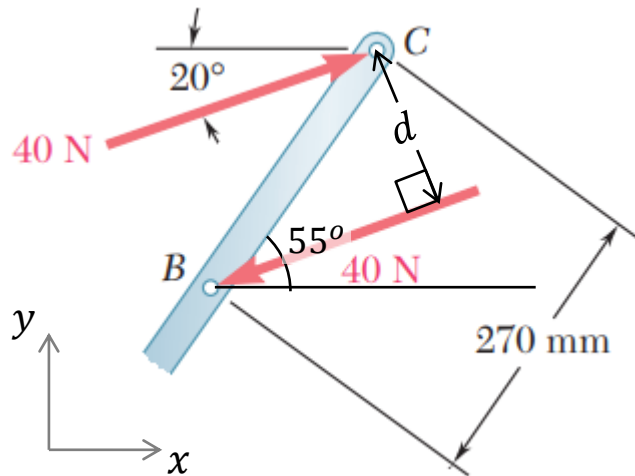
$$\vec{F}_C = -\vec{F}_B = 37.59\vec{i} + 13.68\vec{j}$$

(a) Taking moment about A, gives

$$\begin{aligned} M_A = \vec{r}_{B/A} \times \vec{F}_B + \vec{r}_{C/A} \times \vec{F}_C &= ((0.2237)(-13.68) - (0.3195)(-37.59)) \\ &+ ((0.3786)(13.68) - (0.5406)(37.59)) = -6.195 \text{ N.m} \end{aligned}$$

Example 1.6...

(b) The moment of the couple can be determined by either:



By taking moment about B , we have

$$\begin{aligned}
 M_B &= (r_{C/B})_x F_{C,y} - (r_{C/B})_y F_{C,x} \\
 &= (0.27 \cos 55^\circ)(13.68) - (0.27 \sin 55^\circ)(37.59) \\
 &\Rightarrow M_B = -6.195 \text{ N.m}
 \end{aligned}$$

Finding the perpendicular distance between the forces, i.e.

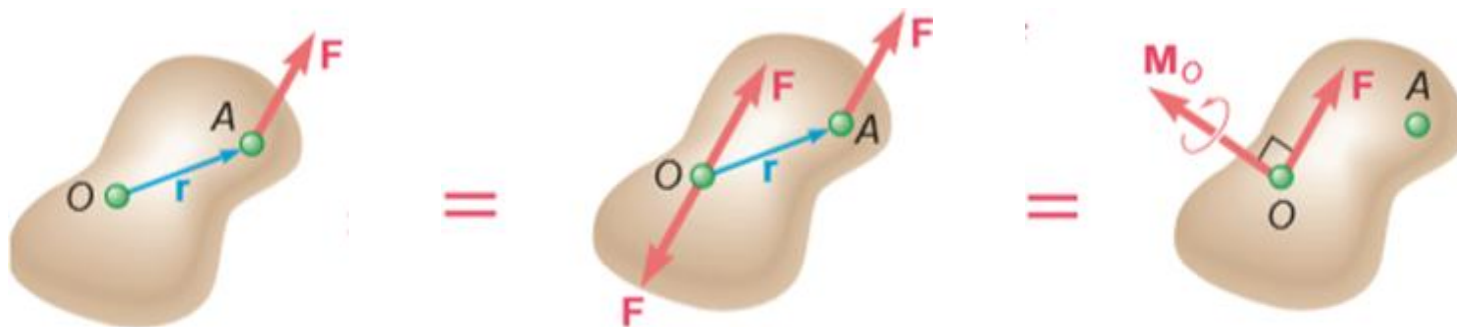
$$d = 0.27 \sin(55^\circ - 20^\circ) = 0.1549 \text{ m}$$

With this, the moment of the couple about B and C is

$$M_B = M_C = -40(0.1549) = -6.195 \text{ N.m}$$

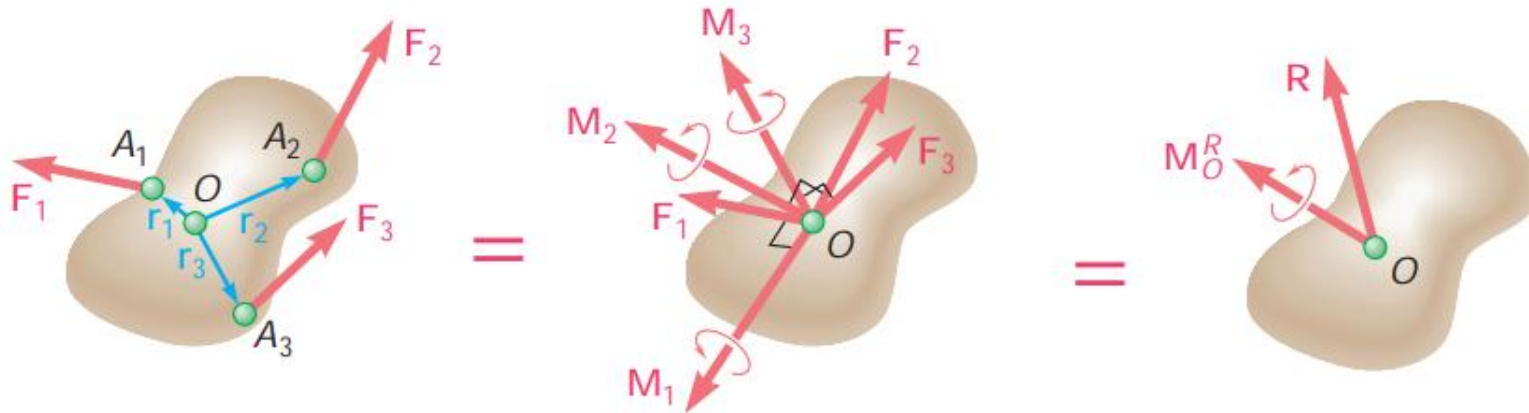
Representing a Force by a Force-Couple Pair

- Consider a force applied on a rigid body. The effect it produces on the rigid body about any point can be represented by an equivalent force-couple pair or system.
 - Attaching an equal and opposite force vectors at O produces no net effect on the body.
 - Finally, the three forces may be replaced by an equivalent force vector and a couple vector, i.e. a force-couple pair. The moment of the couple is given by $\vec{M}_O = \vec{r} \times \vec{F}$, and its direction is perpendicular to the plane containing the moment arm vector $\vec{r}$ and $\vec{F}$.



Reducing a System of Forces to a Force-Couple Pair

- Extending the same concept to a system of forces, one can always reduce the system of forces to an equivalent force-couple pair at any point of reference.



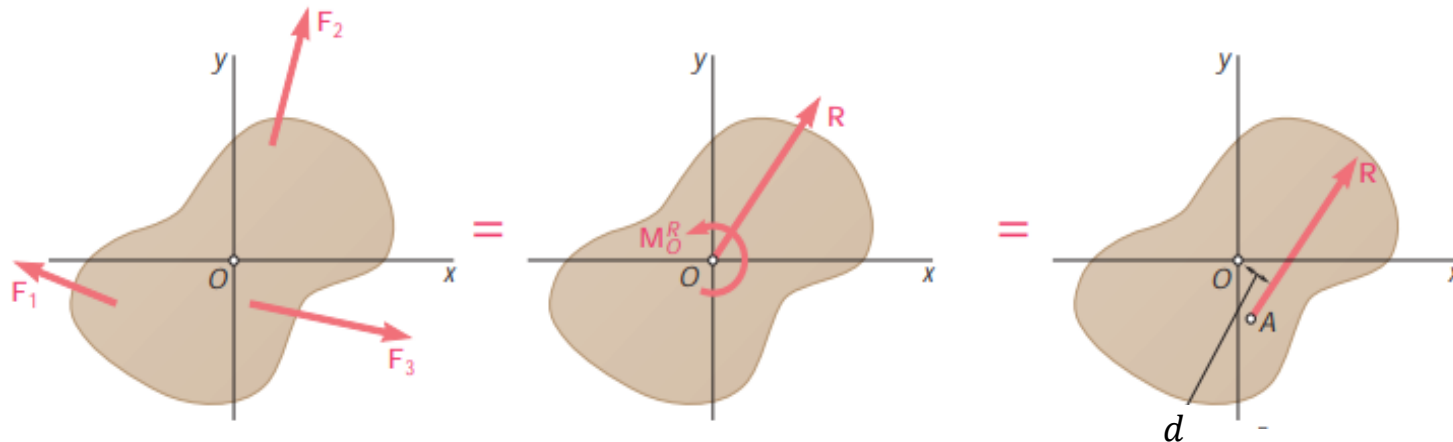
- The equivalent force-couple pair is defined by

$$\vec{R} = \sum \vec{F}_i; \quad \text{and} \quad \vec{M}_O^R = \sum (\vec{r}_i \times \vec{F}_i)$$

- In general, $\vec{R}$ and $\vec{M}_O^R$ are not in the same directions. But for 2D problems, where forces are coplanar, the two resultant vectors will be perpendicular to each others, if they exist.

Representing a Force-Couple Pair by a Force in 2D

- Given that a system of forces can always be reduced to an equivalent force-couple pair $(\vec{R}, \vec{M}_O^R)$ w.r.t. any reference point O . And for 2D planar problems, this force-couple pair is perpendicular to each others.



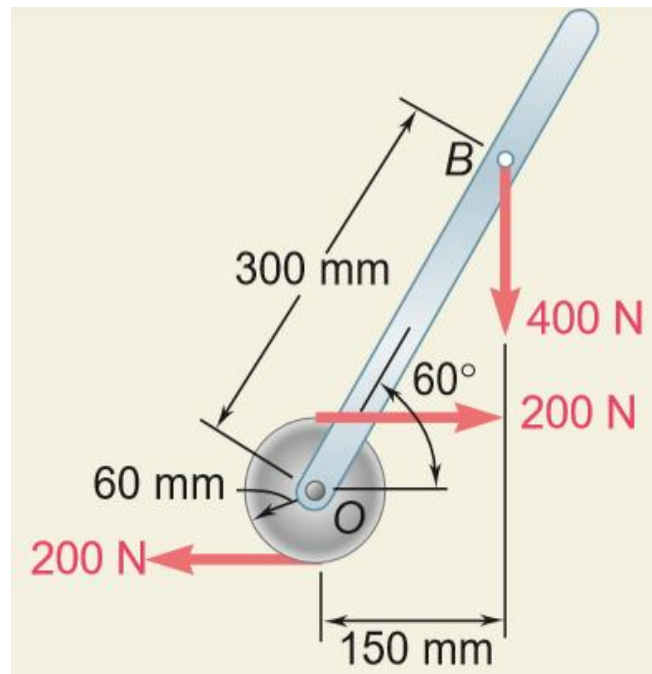
- This force-couple pair can in turns be represented by the resultant force $\vec{R}$, which is shifted perpendicularly to the line-of-action of $\vec{R}$ by a distant given by

$$M_O^R = Rd \Rightarrow d = \frac{M_O^R}{R}$$

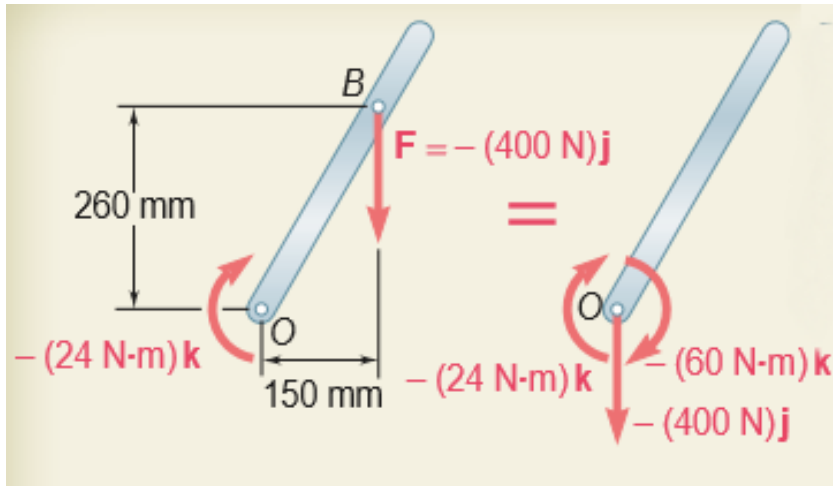
where the sense of the moment $\vec{d} \times \vec{R}$ is identical to $\vec{M}_O^R$.

Example 1.7

Replace the system of forces shown by an equivalent single force applied on the lever. Determine the distance from the shaft at O to the point of application of this equivalent force.



Example 1.7...



Moment due to the 200 N couple is

$$M_{200} = -(200)(0.12) = -24 \text{ N}\cdot\text{m}$$

Moment due to the 400 N force about O is

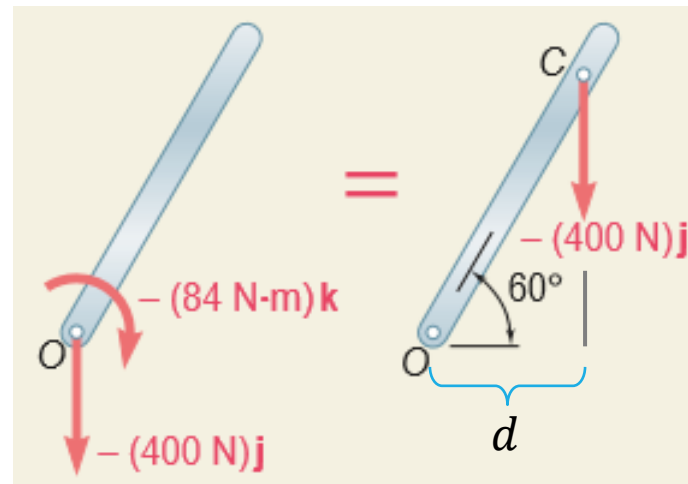
$$M_{400} = -(0.15)(400) = -60 \text{ N}\cdot\text{m}$$

Total moment applied on the lever about O is

$$M_O = M_{200} + M_{400} = -84 \text{ N}\cdot\text{m}$$

Hence, the equivalent force-couple pair at O is -400 N vertical force and 84 N·m clockwise moment.

Example 1.7...



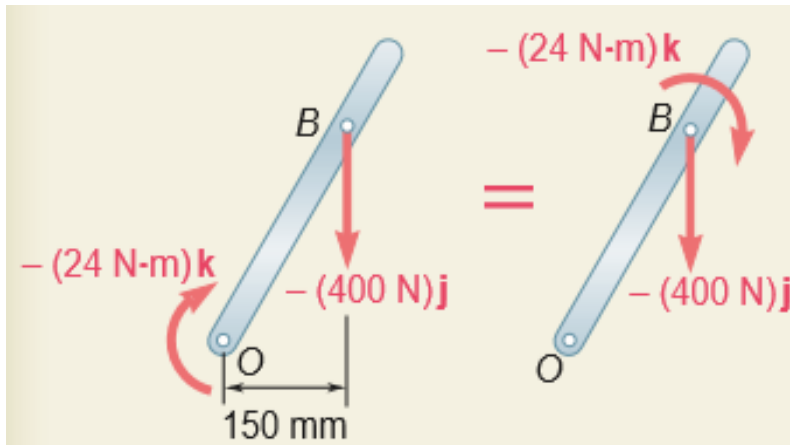
Finally, the vertical force should be shifted to the right of O to produce clockwise moment. And the perpendicular distance to be moved is

$$d = \frac{M_O}{F} = \frac{84}{400} = 0.21 \text{ m}$$

With this, the point C on the lever is

$$OC = \frac{d}{\cos(60^\circ)} = \frac{0.21}{\cos(60^\circ)} = 0.42 \text{ m}$$

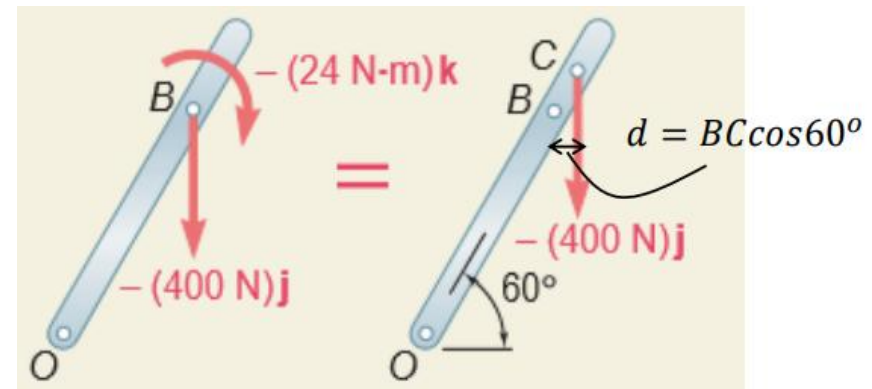
Example 1.7...



Alternatively, we can choose to represent the equivalent force-couple pair at B . In this case, simply move the -24 Nm couple to B , since couple is free moment vector.

Next, we shift the 400 N force to the right to match the 24 Nm moment, i.e.

$$BC \cos 60^\circ = \frac{24}{400} \Rightarrow BC = 0.12$$



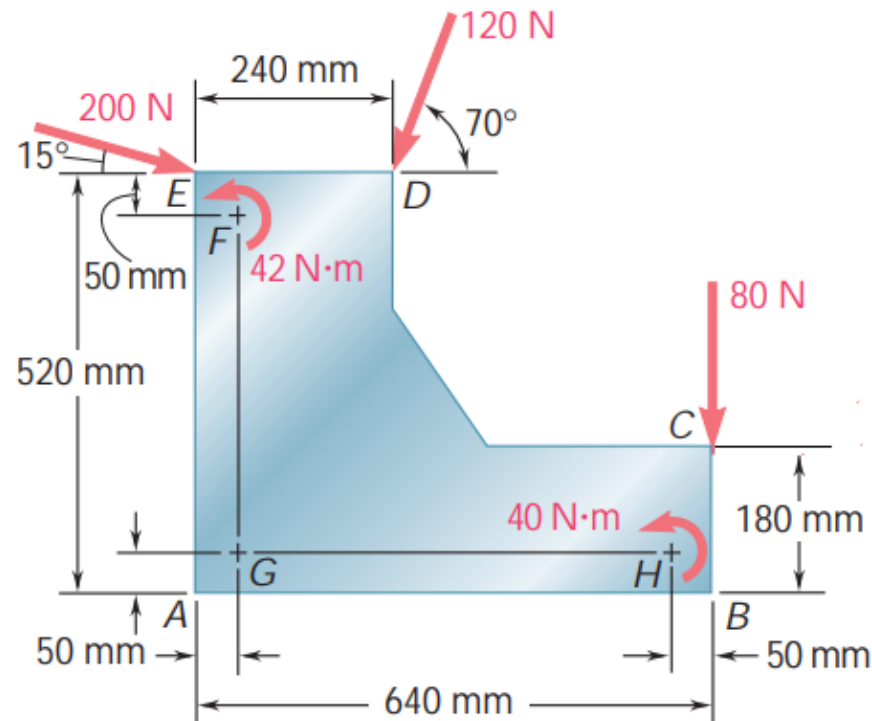
Finally, the position of C with respect to O is

$$OC = OB + BC = \mathbf{0.42 \text{ m}}$$

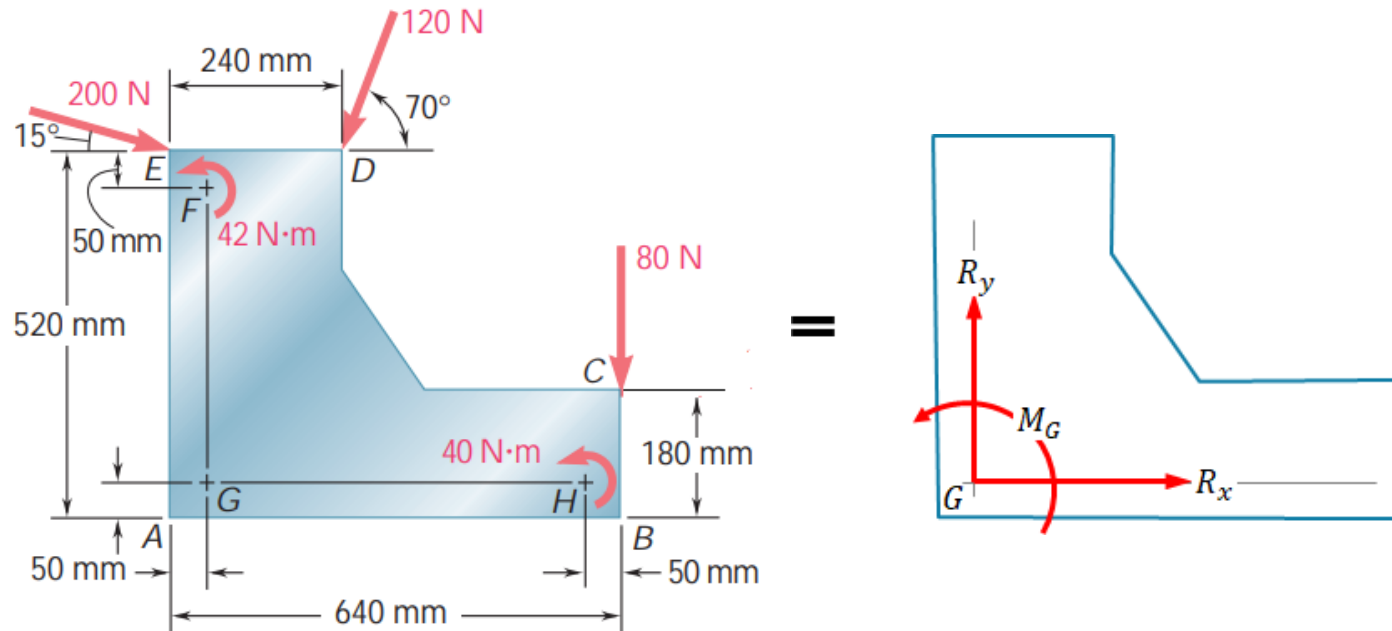
Example 1.8

A machine component is subjected to the forces and couples shown. The component is to be held in place by a single rivet that can resist a force but not a couple. Determine the location of the rivet hole if it is to be located

- (a) on line FG , and
- (b) on line GH .



Example 1.8...



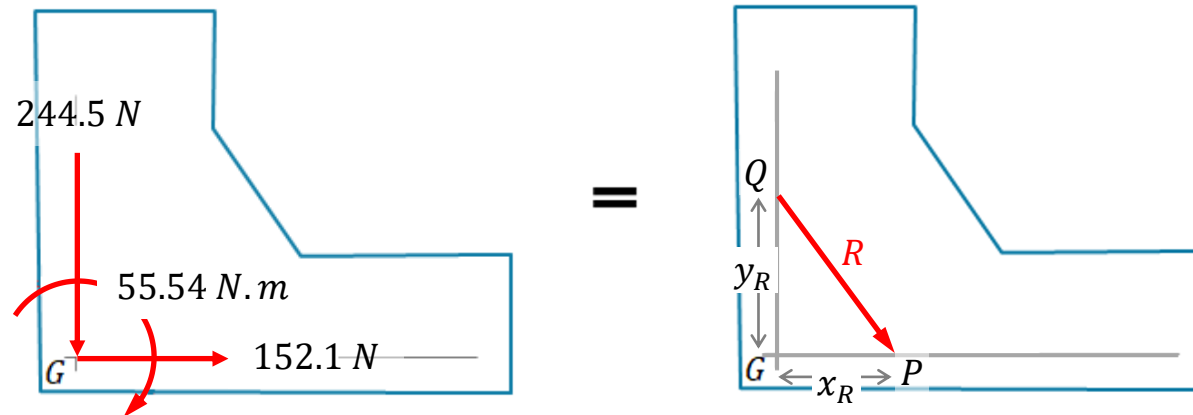
Replacing all the forces and moments with an equivalent force-couple at G, i.e.

$$R_x = 200\cos 15^\circ - 120\cos 70^\circ = 152.1 \text{ N}$$

$$R_y = -200\sin 15^\circ - 120\sin 70^\circ - 80 = -244.5 \text{ N}$$

$$\begin{aligned} M_G = & -(0.47)(200\cos 15^\circ) + (0.05)(200\sin 15^\circ) + (0.47)(120\cos 70^\circ) \\ & - (0.19)(120\sin 70^\circ) - (0.59)(80) + 42 + 40 = -55.54 \text{ N.m} \end{aligned}$$

Example 1.8...



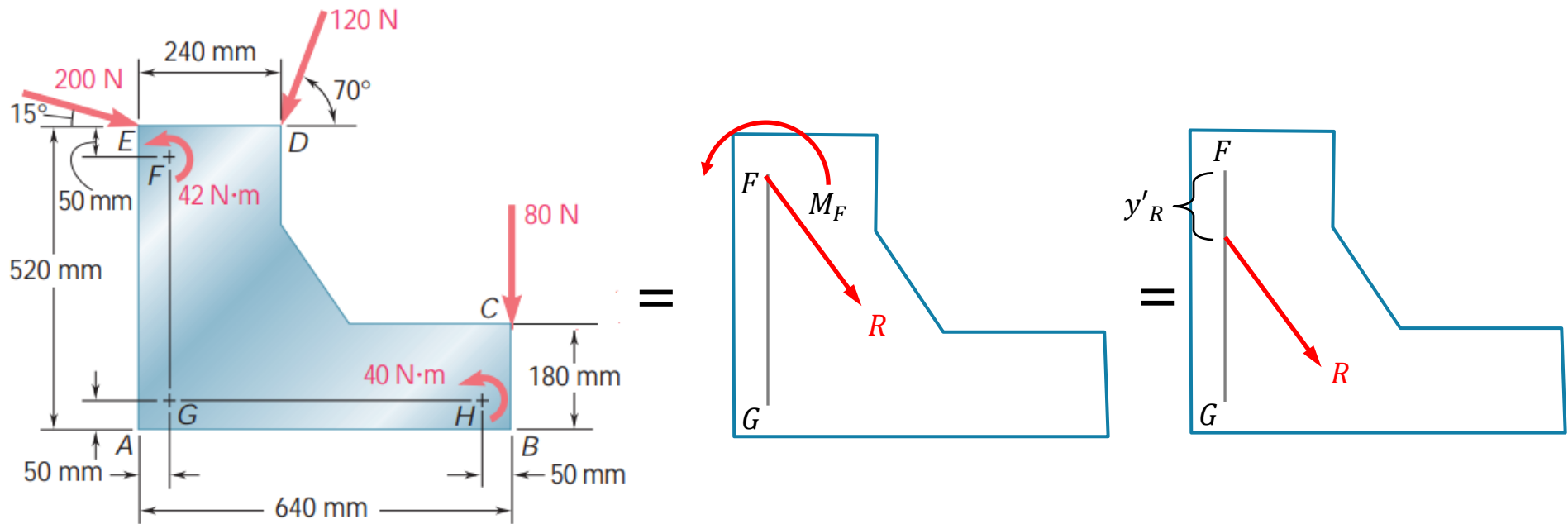
(a) Shifting the equivalent force to x-axis at P, gives

$$x_R = \frac{M_G}{R_y} = \frac{-55.54}{-244.5} = \mathbf{0.227\ m}$$

(b) And moving the force to Q, gives

$$y_R = -\frac{M_G}{R_x} = -\frac{(-55.54)}{152.1} = \mathbf{0.365\ m}$$

Example 1.8...



Suppose the resultant force and moment is calculated at F , which are

$$R_x = 152.1 \text{ N}; R_y = -244.5 \text{ N}; M_F = 8.342 \text{ N}\cdot\text{m}$$

With this coordinate system, the single force would be placed at

$$y'_R = -\frac{M_F}{R_x} = -\frac{(8.342)}{152.1} = -\mathbf{0.0548 \text{ m}}$$